BreatheTwin: how the COPD digital twin works, every morning

1 COLLECT

Wearables and home IoT

Smartwatch: heart rate, breathing rate, steps, sleep
Pulse oximeter: SpO2
Smart inhaler: rescue puffs, doses

Health record (EHR)

Age, GOLD stage, past flare-ups, smoking, cooking smoke, eosinophils, other conditions, BMI, flu vaccine, home oxygen

City air and weather (real)

PM2.5 and AQI from CPCB stations
Temperature, night low, humidity from Open-Meteo
City matched to the patient

2 CLEAN

Daily summary per patient

Phone app / Health Connect / HealthKit / APIs → one row per day → unit and range checks, missing sensor kept as missing (never zero)

3 TWIN

Personal baseline (the twin)

Own normal = median of days t-28 to t-8
Today and 3-day change vs own normal, in own spread

Hybrid risk engine

LightGBM on body, inhaler, weather, record × published PM2.5 dose-response

Rule checks

Breathing +3/min, SpO2 -2 pts, HR +7 bpm vs own normal;
SpO2 below 90% = urgent

4 DECIDE

Flare-up risk in next 1-3 days?

High: call patient today

Medium: watch closely

Low: no action

5 ACT

Ranked call list

Who to phone today, main reason, record facts

Why: reasons

SHAP push of each trigger in plain language

What would help

What-if: purifier, every dose, stay indoors

Smog and cold alerts

Next 3 days per city, with advice

Visit summary

One printable page for the family and doctor

Doctor, nurse or ASHA worker phones the patient or adjusts treatment → outcome recorded → baseline updates the next day

feedback loop